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DEPARTMENT OF COMPUTER ENGINEERING

AI-BASED SCOLIOSIS DETECTION USING DEEP LEARNING

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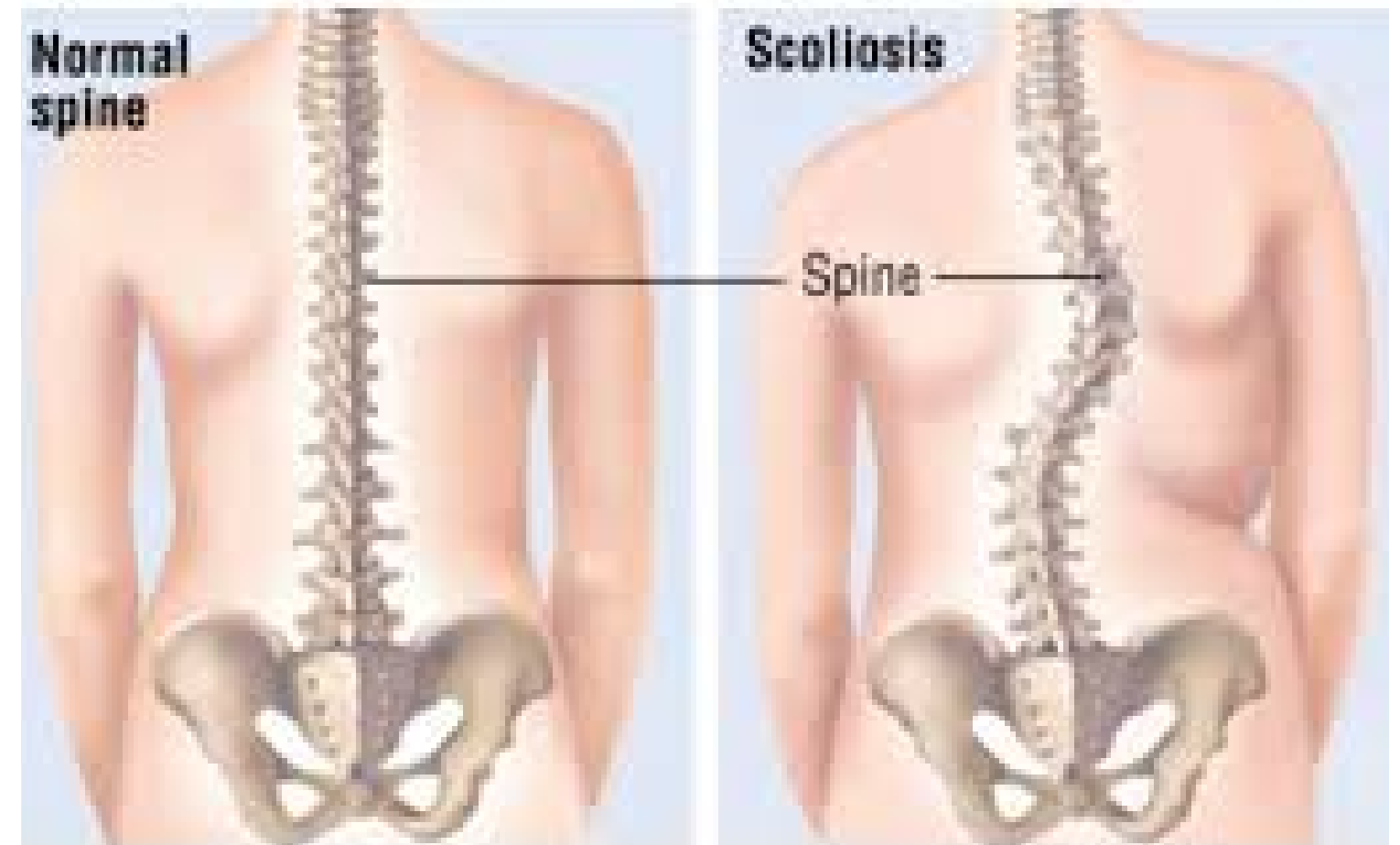
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Problem Statement

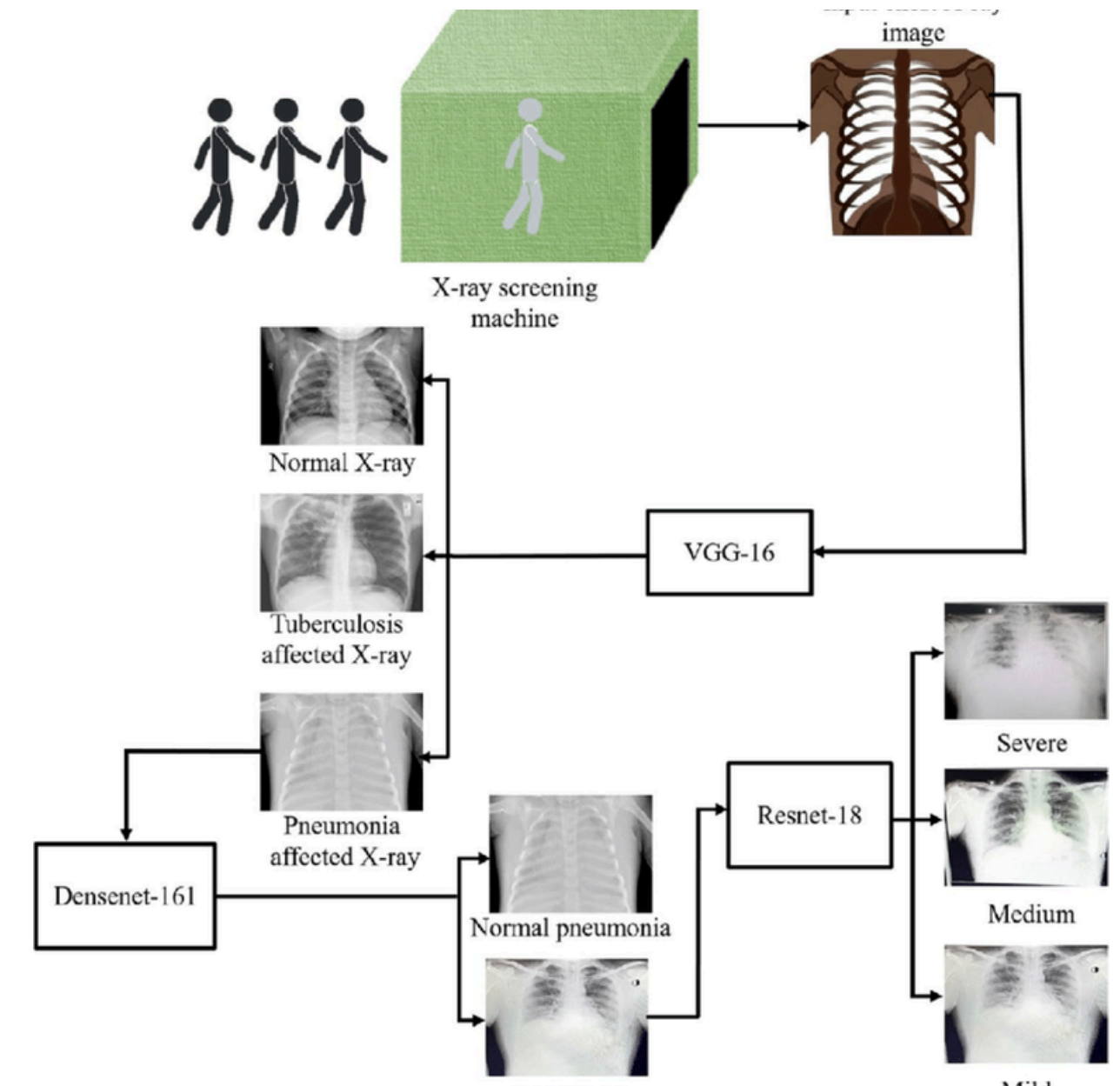
Scoliosis is a medical condition where a person's spine curves sideways in an abnormal "S" or "C" shape.

Early signs of scoliosis in infants are hard to detect with traditional methods. Delayed diagnosis can lead to severe complications. There is a need for a simple, non-invasive system to detect scoliosis early and accurately.



Objectives

- Develop a Convolutional Neural Network (CNN): To classify spinal X-ray images into two categories: Normal or Scoliosis with high accuracy.
- Design a User-Friendly Interface: Implement a Streamlit-based web application for real-time testing and demonstration of the model.
- Enhance Accessibility in Resource-Limited Areas: Enable early and remote diagnosis, especially in under-resourced and rural healthcare settings.



DataSet & Preprocessing

- Source: Mendeley Data (2021)
- Total Images: 259
 - 188 – Scoliosis cases
 - 71 – Normal cases
- Format: Frontal X-ray images
- Type: Labeled data for binary classification
- Challenge: Small size and class imbalance (72% scoliosis)

Preprocessing Steps

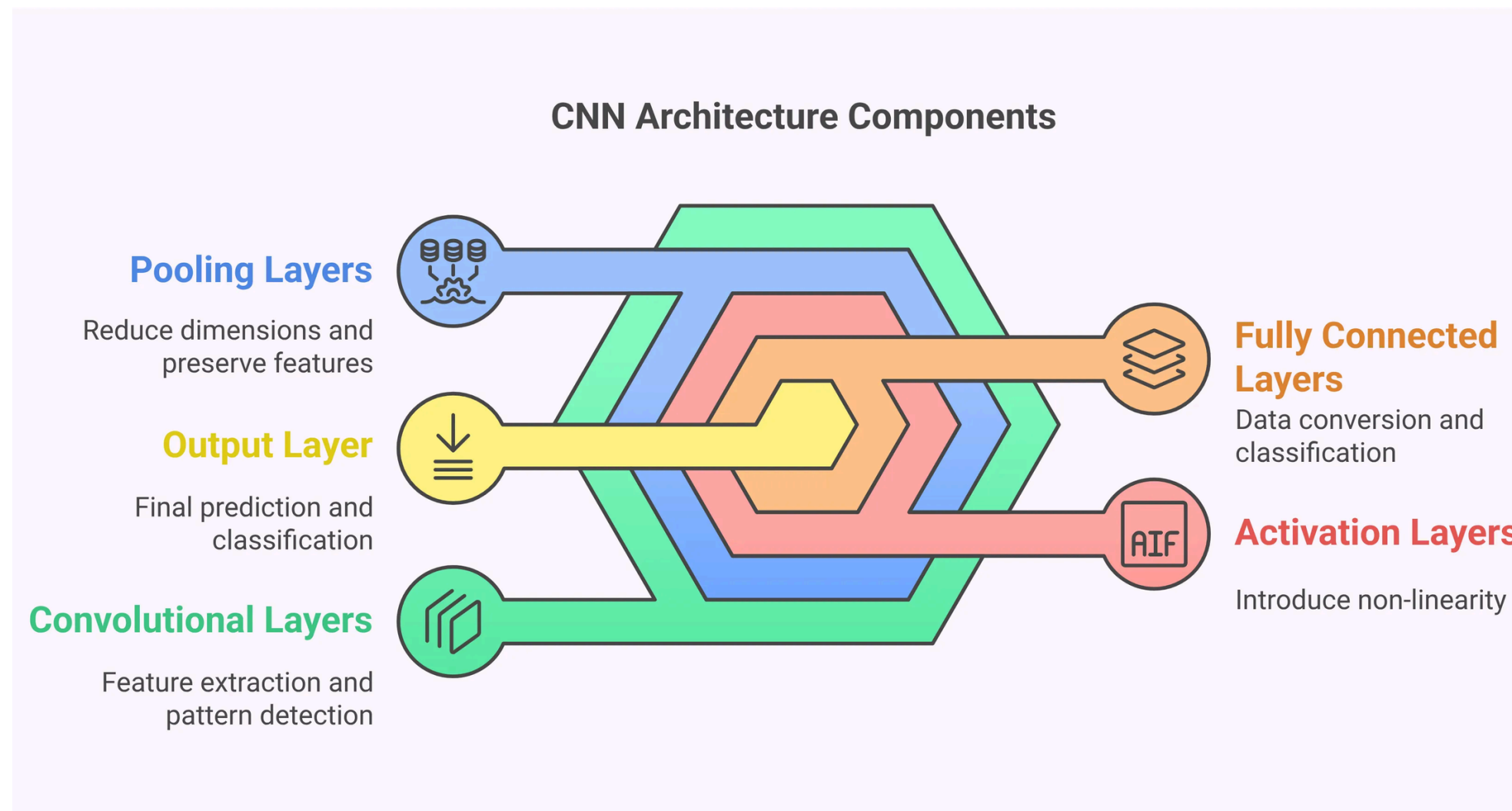
- Resizing to 224×224 pixels
 - Ensures uniform input size for the CNN
- Pixel Normalization
 - Scales values to 0–1 for better training performance
- Label Encoding
 - Converts 'Normal' and 'Scoliosis' to binary (0 and 1)

CNN Model Architecture

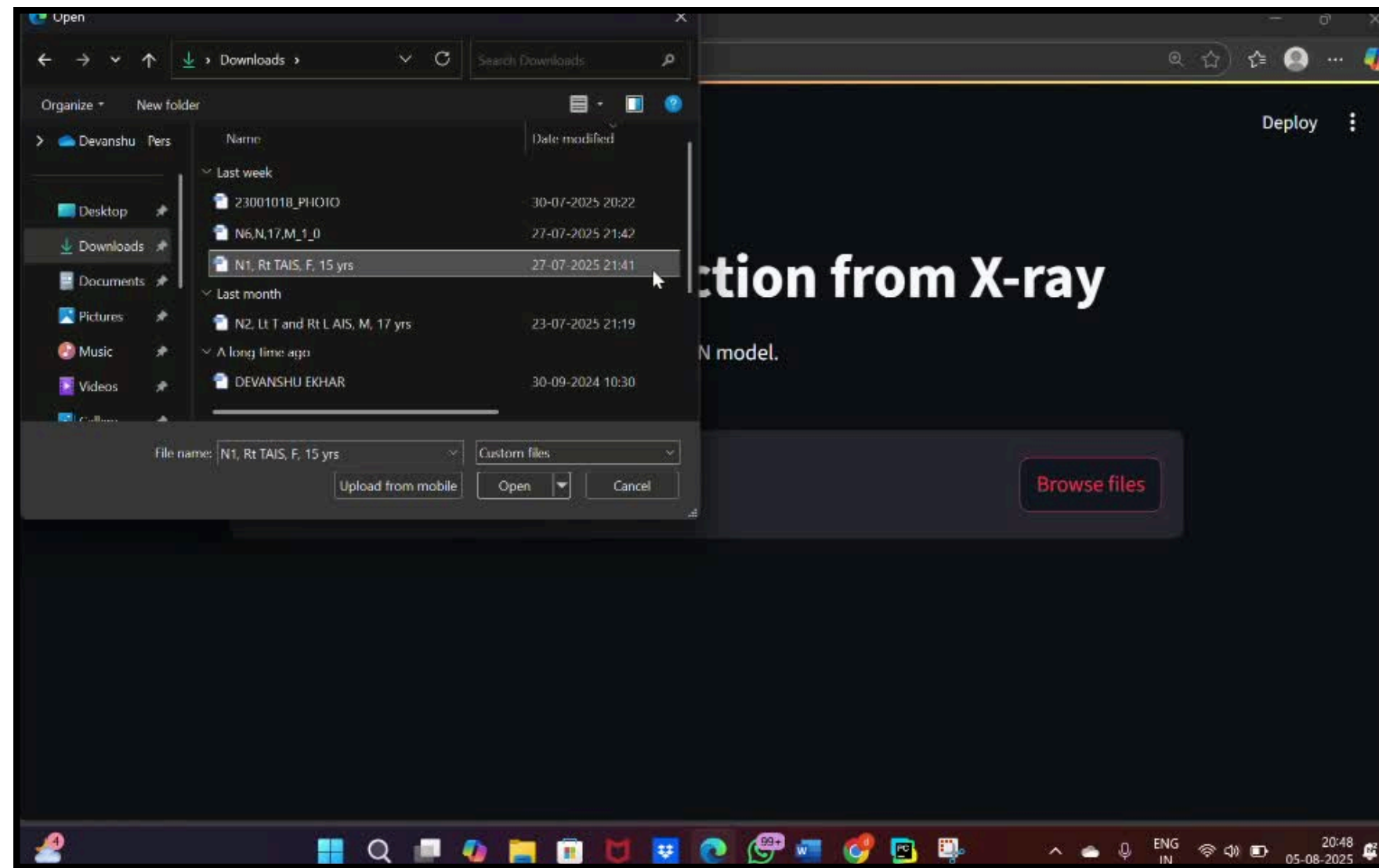
- Input Layer: 224×224 grayscale image
- Convolutional Layers: Feature extraction
- Pooling Layers: Downsampling
- Dense Layers: Classification
- Dropout Layer: Reduces overfitting
- Output Layer: Binary classification (Scoliosis / Normal)

Training Details:

- Loss Function: Binary Cross-Entropy
- Optimizer: Adam
- Epochs: 10
- Frameworks: TensorFlow/Keras

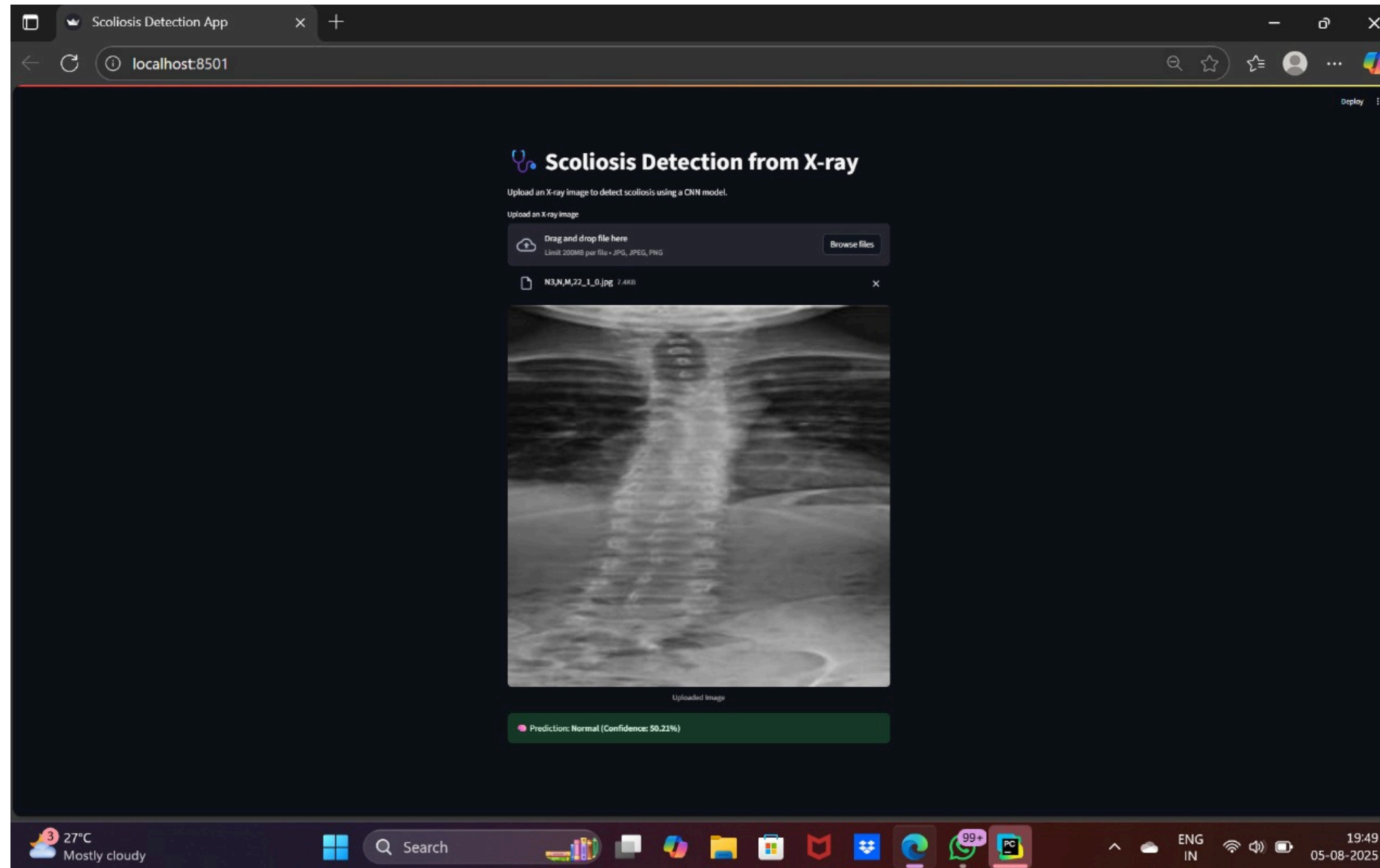


Model Performance



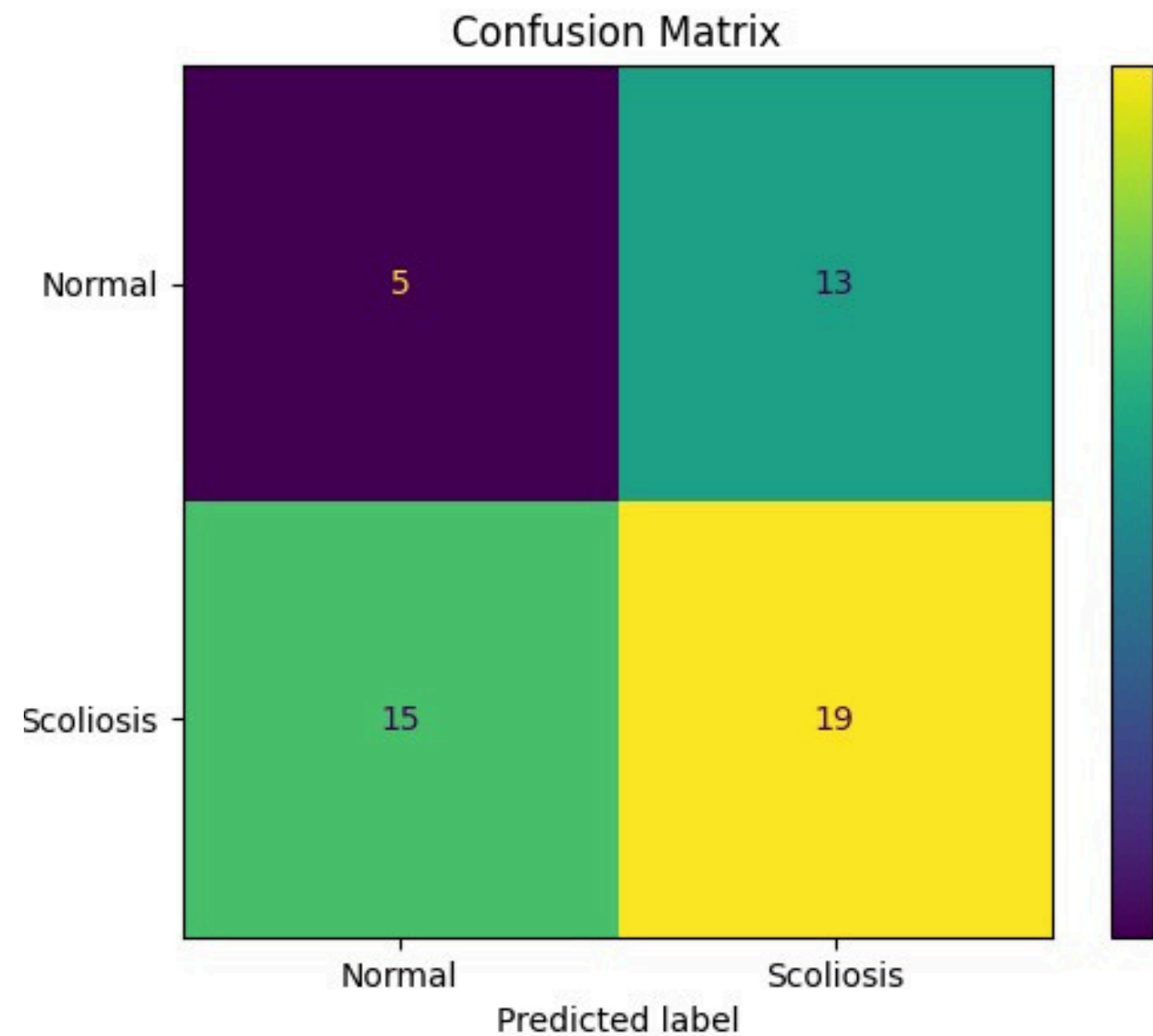
- Training Accuracy: 88.65%
- Validation Accuracy: 78.85%
- Loss Decreased Across Epochs: Indicates learning and convergence
- Insights:
 - Good performance with a small dataset
 - Slight overfitting controlled with dropout

Implementation



- Developed using Streamlit
- Lightweight and accessible interface
- User Actions:
Upload .jpg or .png X-ray
Receive prediction instantly
- No technical expertise required

Confusion Matrix Analysis



- Evaluates classification accuracy

Metrics:

- True Positive (TP): Correctly predicted scoliosis
- True Negative (TN): Correctly predicted normal
- False Positive (FP): Normal classified as scoliosis
- False Negative (FN): Scoliosis classified as normal
- Model performs better at detecting scoliosis

Limitations

- Small Dataset: Only 259 images limits generalization
- Binary Output: No severity classification or curve type
- 2D X-rays Only: Cannot assess spinal rotation or 3D deformity
- No Clinical Validation Yet: Needs real-world hospital testing
- User-uploaded Image Quality: Poor input reduces accuracy

Future Enhancements

- Expand Dataset: More images, diverse age groups
- Multi-Class Model: Add severity detection and curve localization
- 3D Reconstruction: Use lateral X-rays or CT scans
- Mobile/Edge Deployment: For rural use without internet
- EHR Integration: Connect with hospital databases

Conclusion

- CNN-based deep learning demonstrates strong potential for automated scoliosis screening, achieved strong performance on limited data.
- Enables early detection, especially in remote or underserved regions.
- With further enhancement, it can support AI-driven diagnosis in healthcare.

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